**Physics Lab 5** (2nd April 2025):

# The Quantisation of Matter

## The Franck-Hertz experiment

*by* **Mayad Mnif** *and* **Toma Niculica**

---

## Contents

**1 Introduction** **2**

1.1 Objectives/Outline . . . . . 2

1.2 Preparation Work . . . . . 2

**2 Qualitative observations** **4**

2.1 Preliminary tasks - Shape of the Potential Energy . . . . . 4

2.2 Experimental Franck-Hertz curve . . . . . 6

**3 Quantitative measurements** **6**

3.1 Data acquisition with a digital oscilloscope . . . . . 6

3.2 Determination of the excitation energy of the Hg atom . . . . . 7

3.3 Interpretation . . . . . 8

**4 Letter** **9**

**5 Varying the T parameter** **9**

# 1 Introduction

## 1.1 Objectives/Outline

The Franck-Hertz experiment, first performed in 1914 by James Franck and Gustav Hertz, was a pivotal moment in the development of quantum physics. It provided the first direct evidence that atoms absorb energy in discrete amounts — a foundational idea in quantum theory. Their work supported Niels Bohr's model of the atom, which proposed that electrons transition between fixed energy levels, absorbing or emitting energy in quantized steps.

In this lab session, we aim to replicate the Franck-Hertz experiment to:

- Demonstrate that atoms can be excited only by specific, quantized amounts of energy.
- Observe that electrons lose energy in distinct steps as they interact with mercury atoms.
- Determine the excitation energy of mercury from the experimental data.

## 1.2 Preparation Work

1. The force $\mathbf{F}$ exerted on a particle with charge $q$ in an electric field $\mathbf{E}$ is given by:

$$\mathbf{F} = q\mathbf{E} \tag{1}$$

   For an electron, $q = -e$, where $e$ is the elementary charge. Therefore, the force on an electron is:

$$\mathbf{F} = -e\mathbf{E} \tag{2}$$

   This indicates that the force on the electron is in the direction opposite to the electric field.

2. In a one-dimensional electrostatic system where the potential depends only on the $x$-direction, and assuming a uniform setup between two electrodes (anode and cathode), the electric field $E_x$ between the electrodes is constant:

$$E_x = \text{constant} \tag{3}$$

The field points from the anode (higher potential) to the cathode (lower potential).

3. The electric field $E_x$ in terms of the electric potential $V$ is given by:

$$E_x = -\frac{dV}{dx} \tag{4}$$

Assuming a linear potential drop between the cathode and anode separated by a distance $d$, with a potential difference $\Delta U = V_{\text{anode}} - V_{\text{cathode}} > 0$, the electric field is:

$$E_x = -\frac{\Delta U}{d} \tag{5}$$

This signifies a uniform electric field directed from the anode to the cathode.

4. The potential energy $E(x)$ of an electron in an electric potential $V(x)$ is:

$$E(x) = -eV(x) \tag{6}$$

Assuming a linear potential $V(x)$ increasing from the cathode to the anode, the potential energy $E(x)$ decreases linearly along the $x$-axis.

5. The kinetic energy $KE$ gained by an electron moving through a potential difference $\Delta U$ is equal to the work done on the electron:

$$KE = e\Delta U \tag{7}$$

Expressed in electron-volts (eV), where $1\,\text{eV} = 1.602 \times 10^{-19}\,\text{J}$:

- For $\Delta U = 1\,\text{V}$:

$$KE = 1\,\text{eV} \tag{8}$$

- For $\Delta U = 100\,\mathrm{V}$:

$$KE = 100\,\mathrm{eV} \tag{9}$$

# 2 Qualitative observations

## 2.1 Preliminary tasks - Shape of the Potential Energy

1. **Potential Energy Profile of the Electron**

   Assuming $U_A = 50\,\mathrm{V}$ and $U_E = 1\,\mathrm{V}$, the potential energy profile $E(x) = -eV(x)$ of the electron as it moves inside the Franck-Hertz tube is shown below. The end displays a short rise which happens due to $U_E$ to limit the noise caused by low-energy electrons that may reach the anode.

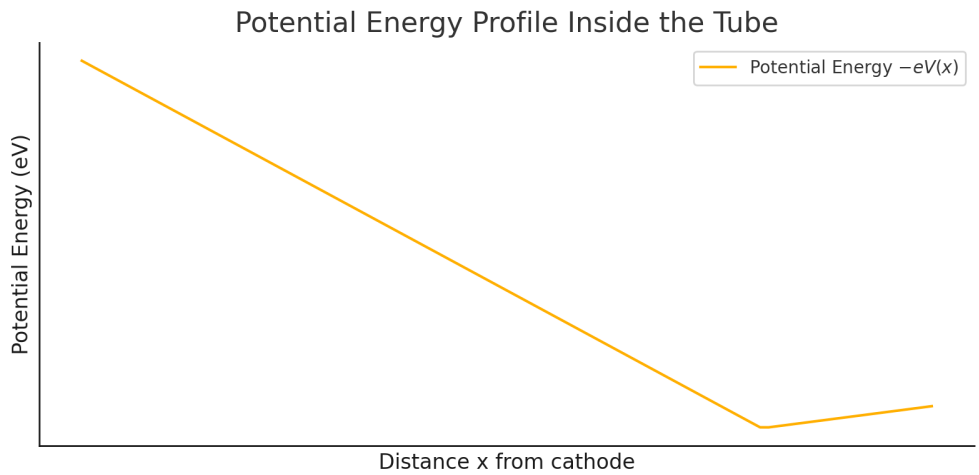


Figure 1: Schematic of the potential energy $-eV(x)$ inside the tube.

2. **Kinetic Energy at the Grid Without Collisions**

   If no collisions occur with Hg atoms, then energy is conserved. At the start, let the potential energy be $E_p$, which is also the total energy as the kinetic energy is 0 at rest. When it reaches the grid, the kinetic energy will be the total energy minus the potential energy, thus we have:

$$KE = E_p - (-eU_A)$$

   Furthermore, we can consider $E_p = 0$ by definition.

   Finally,

$$KE = eU_A \quad \Rightarrow \quad KE = U_A\ (eV)$$

3. **Minimum Kinetic Energy to Reach the Anode**

   For the electron to be able to reach the anode, its kinetic energy must be equal to at least the work required to get to the anode, which is $eU_E$.

   Hence we must have:

$$E_{k,\text{min}} = eU_E \quad \Rightarrow \quad E_{k,\text{min}} = U_E\ (eV)$$

4. **Why Elastic Collisions Are Negligible**

   An elastic collision conserves both energy and linear momentum and the solution to the those two equations results in the following approximation given in the appendix:

$$\frac{\Delta E_K}{E_K} \approx -\frac{4m}{M}$$

   Given $\frac{m}{M} \approx \frac{1}{200000}$, the loss is negligible. In contrast, inelastic collisions are by definition collisions where energy is not conserved

5. $I = f(U_A)$ **Qualitative dependence**

   The presence of inelastic collisions makes it tougher for electrons to reach the anode, requiring more accelerating voltage for the same intensity. Thus, the curve will be shifted to the right.

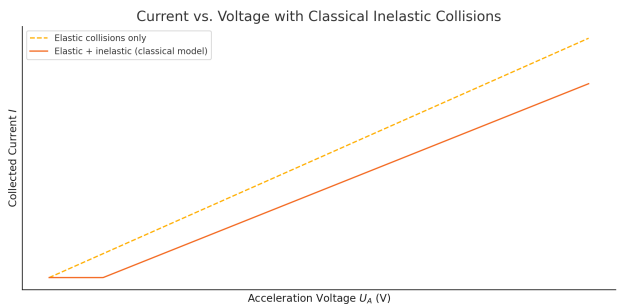


Figure 2: Fig. 2 Elastic and Inelastic collisions

## 2.2 Experimental Franck-Hertz curve

Sweeping the value of $U_A$ from 0 to around 50V (staying in the safe range) we observed the sudden decrease of the y-value ($I \propto U_y$) followed by a rise back up at certain x values (accelerating voltage). We estimated that they occur around the following values of $U_A$: 20,30,38,47. This experimental observation directly contradict our idea presented above where we constructed the pseudo-linear current-voltage graph, since its underlying assumption is that the atom follows a planetary model of continous energy levels. In reality, however, due to the discrete nature of energy, mercury atoms only get excited at certain energy levels, which are a multiple of the first one. Our measurement by hand is very imprecise, since altough we noted maximas and minimas down as we played with the voltage, we could have easily skipped extrema or mislocated them.

# 3 Quantitative measurements

## 3.1 Data acquisition with a digital oscilloscope

In this part, we acquired the Franck-Hertz curve in an automatized fashion thanks to a digital oscilloscope. This allowed us to make reliable and quantitative measurements and extract precisely the structure of the energy levels of the Hg atom. We used the oscilloscope's "ramping mode", which provides a continuous variation of the accelerating voltage to have a smooth and high-resolution dataset, revealing the precise minima and maxima on the Franck-Hertz curve. We had to select carefully the resolution of the oscilloscope in order to obtain a detailed curve where we can identify the inelastic collisions without measuring too much noise. Here is the curve that we obtained for the Franck-Hertz experiment at 200°C.

The Franck-Hertz curve exhibits distinct minima and maxima separated by a pseudo-periodic voltage interval, corresponding to the discrete energy loss of electrons during inelastic collisions with mercury atoms, which reveals the quantized excitation energy of Hg.

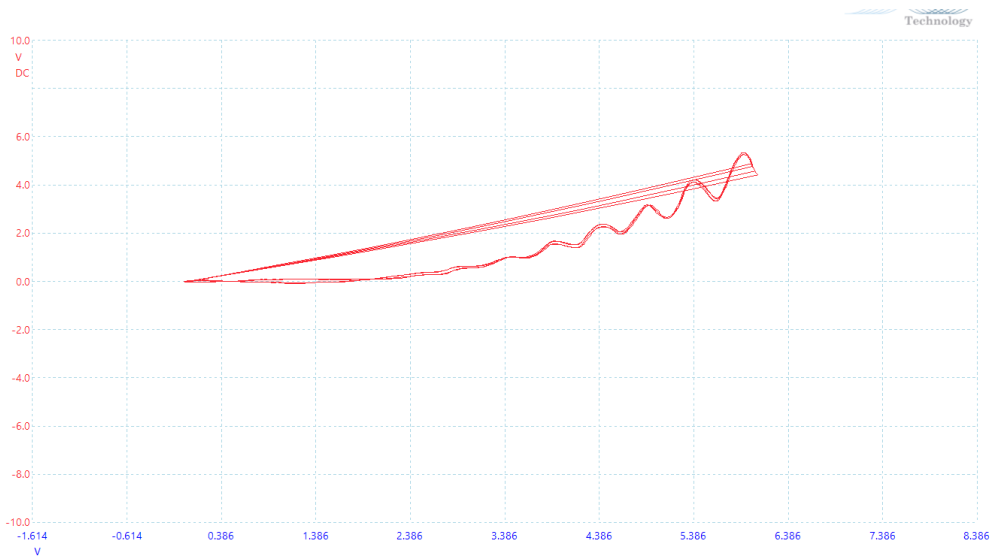


Figure 3: Fig. 3 Franck-Hertz curve ($U_y$ as a function of $U_x$)

## 3.2 Determination of the excitation energy of the Hg atom

Here is the table we extracted from the Franck-Hertz curve, with the successive minima and maxima of the curve. We computed the excitation energy of the Hg atom by calculating the difference of voltage between the "valleys" and multiplying by $e$. After taking into account the $1/10$ factor from the voltage output we chose, we find an excitation energy for the Hg atom of $(7.87 \pm 0.03) \times 10^{-19}$ J, or $4.92 \pm 0.02$ eV.

Table 1: Franck-Hertz Data

| $n$ | $U_{\text{min}}$ (V) | $U_{\text{max}}$ (V) | $\Delta U_{\text{min}}$(V) | $\Delta E$ (J) |
|---|---|---|---|---|
| 1 | - | 2.47 | - | - |
| 2 | 2.68 | 2.92 | 0.46 | $7.36 \times 10^{-20}$ |
| 3 | 3.14 | 3.44 | 0.48 | $7.68 \times 10^{-20}$ |
| 4 | 3.62 | 3.89 | 0.52 | $8.32 \times 10^{-20}$ |
| 5 | 4.14 | 4.41 | 0.48 | $7.68 \times 10^{-20}$ |
| 6 | 4.62 | 4.89 | 0.5 | $8.00 \times 10^{-20}$ |
| 7 | 5.12 | 5.38 | 0.51 | $8.16 \times 10^{-20}$ |
| 8 | 5.63 | 5.9 | - | - |

All errors presented above were calculating using the standard deviation formula given in the lab presentation.

## 3.3 Interpretation

The Franck-Hertz curve exhibits periodic minima at voltages $U_A \approx 4.9\,\text{V}, 9.8\,\text{V}, \ldots$, corresponding to electrons losing discrete amounts of energy to Hg atoms. From the data:

- Mean excitation energy: $\Delta E = (7.87 \pm 0.03) \times 10^{-19}\,\text{J} \approx 4.92\,\text{eV}$.
- This matches the first excitation energy of Hg, corresponding to the transition:

$$6^1S_0\,(\text{ground state}) \rightarrow 6^3P_1\,(\text{first excited state}).$$

Electrons accelerated to 4.9 V gain 4.9 eV, sufficient to excite Hg atoms to $^3P_1$.

When Hg atoms de-excite from $6^3P_1$ to $6^1S_0$:

- Photon energy: $\Delta E = 4.9\,\text{eV}$.
- Wavelength:

$$\lambda = \frac{hc}{\Delta E} = \frac{1240\,\text{eV}\cdot\text{nm}}{4.9\,\text{eV}} \approx 253\,\text{nm}\,(\text{ultraviolet}).$$

While the primary transition emits UV (253 nm), we observed visible blue light ($\lambda \approx 435\,\text{nm}$), which likely arises from secondary transitions between higher energy states, such as:

$$7^3S_1 \rightarrow 6^3P_1\,(\text{blue},\ \lambda \approx 435\,\text{nm}).$$

The Franck-Hertz experiment demonstrates that mercury atoms can indeed be excited to multiple quantum states through electron collisions. The clear periodic minima in the current-voltage curve, at regular intervals of approximately 4.9 eV, provide direct evidence of quantized energy absorption. Moreover, the observation of blue light emission from the tube shows us that higher-energy excitations (at all multiples of the excitation energy 4.9eV) are achievable. These results demonstrate how electrons can transfer discrete amounts of energy to mercury atoms.

# 4 Letter

Dear Professor Bohr,

I hope this letter finds you well. I'm writing to share exciting results from Prof. Hertz and I's recent experiment with mercury vapor. We observed clear, regular drops in electron current corresponding to energy losses of 4.9 eV, a confirmation of your atomic model's prediction of quantized energy levels. The periodic pattern in our measurements suggests electrons are transferring discrete packets of energy to the mercury atoms. These findings seem to provide strong experimental support for your ideas about atomic structure. I would be grateful for your thoughts on how this might further refine our understanding of quantum states.

Best regards,

James Franck

# 5 Varying the T parameter

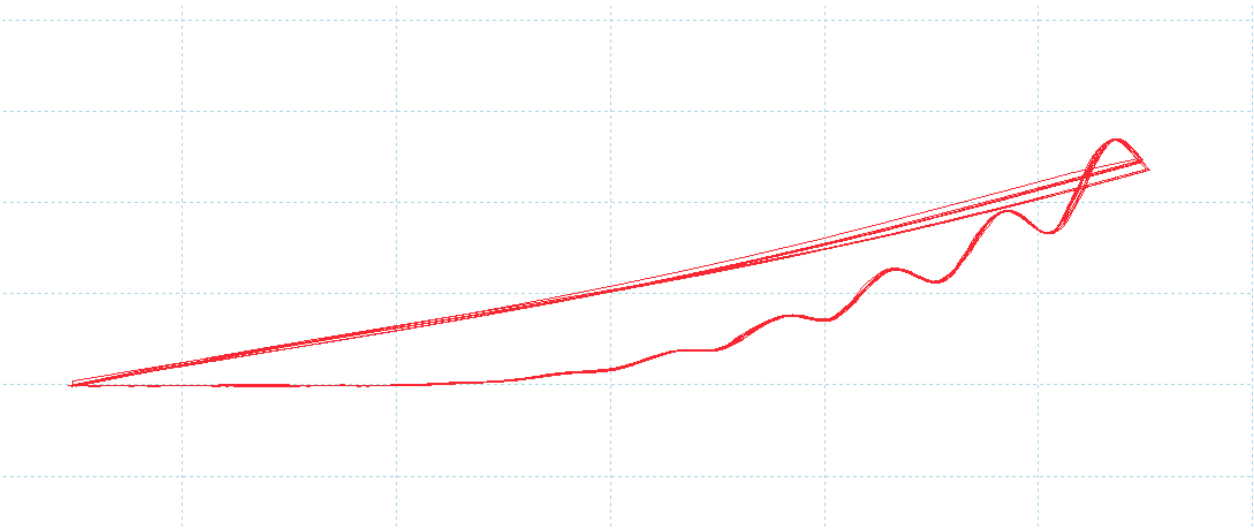

Figure 4: Franck-Hertz curve - T = 180 C

Figure 5: Franck-Hertz curve - T = 160 C

After computing the mean energy interval (distance between minimas) for each temperature, we used Planck's formula again to get the corresponding wavelength.

We have thus got:

$$\lambda_{180} \approx 236 \text{ nm} \quad \lambda_{160} \approx 226 \text{ nm}$$

Notice that temperature thus affect the wavelength corresponding to the excitation energy levels. Wavelength looks to increase as temperature increasing, so the energy required goes down as a function of temperature.

This could be possibly due to changes in the mean free path of electrons, atomic density, or collisional broadening effects. However, no definitive theoretical model currently explains this behavior in the context of the Franck-Hertz experiment, and further investigation would be required to confirm and quantify this relationship